Confinement - Deconfinement Phase Transition in SU(2) LGTs

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Background

The Anatomy of Matter

Zooming in

System	**vs.**	**Strong Force (QCD)**
Hydrogen Atom	⟵ **Scale** ⟶	Proton / Neutron
Electromagnetism	⟵ **Force** ⟶	Strong Nuclear Force
Electric Charge	⟵ **Charge** ⟶	Color Charge (R, G, B)
Photons (γ)	⟵ **Mediator** ⟶	8 Gluons (g)

The Confinement Paradox

Quarks are "trapped" — we cannot isolate them to study them directly.

Goal: Simplify $SU(3) \to SU(2)$ to study gluon-only dynamics.

The Dual Nature of QCD

The coupling α_s runs with energy, creating two distinct regimes:

Asymptotic Freedom

High Energy / Short Distance

- $\alpha_s \ll 1$
- Quarks behave as free particles
- ✓ **Solvable perturbatively**

Confinement

Low Energy / Long Distance

- $\alpha_s \sim \mathcal{O}(1)$
- Quarks are permanently bound
- ✗ **Perturbation theory fails**

Conclusion: We must build a discrete universe inside a computer.

The Bridge: Quantum to Statistical

1. Path Integral

$$Z = \int \mathscr{D}A \ \exp(iS)$$

Problem: Rapid oscillations make direct numerical integration impossible.

2. Wick Rotation $(t \to -i\tau)$

$$Z = \int \mathscr{D}\phi \ e^{-S_E[\phi]}$$

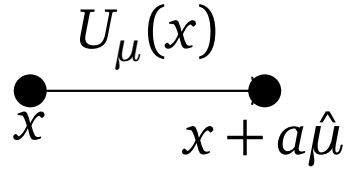
Solution: e^{-S_E} is a positive weight.
The vacuum becomes a **Thermal Fluid**.

Quantum Action S_E $\equiv$ Statistical Energy

Wilson's Solution: Link Variables

Discretising spacetime requires preserving **local gauge symmetry**.

- **Sites:** Where quarks would live.
- **Links (U_μ):** Gluon fields as group elements.

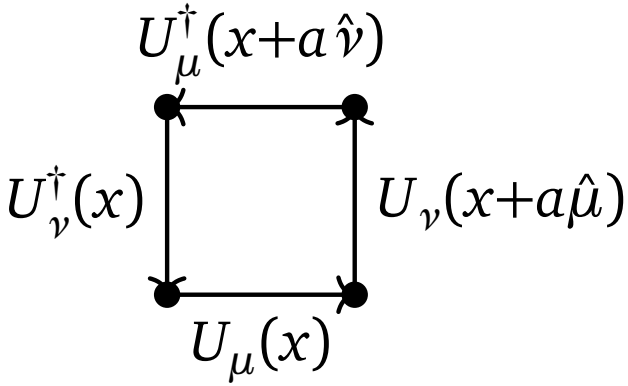


$$U_\mu(x) = \exp\big(igaA_\mu(x)\big) \in SU(2)$$

- **Gauge Invariance:** Links act as "translators" between adjacent sites.

The Wilson Action: The "Energy Cost"

The simplest gauge-invariant quantity is the **Plaquette** $U_\square$: a 1×1 closed loop.



$$U_\square = U_\mu U_\nu U_\mu^\dagger U_\nu^\dagger$$

$$S_\square = \beta\left(1 - \tfrac{1}{2}\,\mathrm{Tr}\,U_\square\right)$$

- $\beta = 4/g^2$: inverse coupling constant.
- Gauge symmetry preserved at every lattice spacing.

Observable I: The Staple

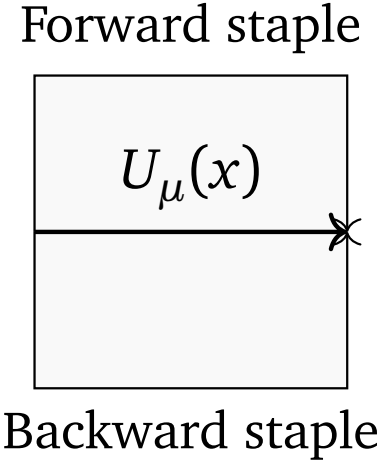
Definition

The sum of "forward" and "backward" plaquettes in all directions $\nu \neq \mu$:

$$K_\mu(x) = \sum_{\nu \neq \mu} \Big[U_\nu(x)\, U_\mu(x+\hat{\nu})\, U_\nu^\dagger(x+\hat{\mu}) + U_\nu^\dagger(x-\hat{\nu})\, U_\mu(x-\hat{\nu})\, U_\nu(x-\hat{\nu}+\hat{\mu}) \Big]$$

Physical Meaning:

- Encodes the **local gluonic field strength** around a link.
- In the continuum limit, approximates the field tensor $F_{\mu\nu}$.



Boltzmann Weight

The local action change

$$\Delta S_{\square} = -\beta \left(\frac{1}{2} \mathbb{R} \, \mathrm{Tr} \left(U_{\mu} K_{\mu} \right) \right)$$

and the probability

$$P(U) \propto exp(-S_{\square})$$

Observable II: The Polyakov Loop

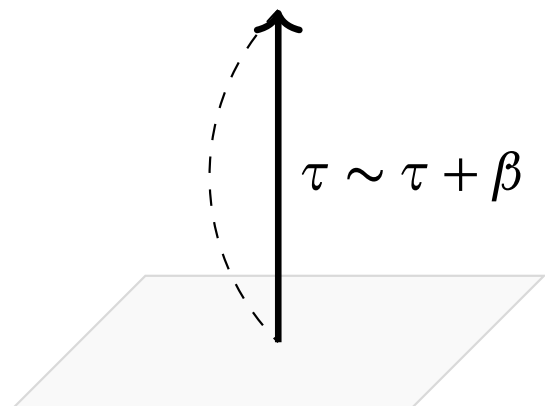
Definition

A path wrapping around the periodic Euclidean time (τ) dimension:

$$L(\vec{x}) = \frac{1}{2}\,\mathrm{Tr}\prod_{t=1}^{N_\tau} U_0(\vec{x}, t)$$

Physical Meaning:

- Measures the free energy of a single static quark: $\langle L \rangle \sim e^{-F/T}$.
- **Confinement:** $\langle L \rangle = 0$ (infinite energy).
- **Deconfinement:** $\langle L \rangle \neq 0$ (finite energy).



Implementation

Comprehending the scale

- A tiny $12^3 \times 4$ lattice has $\sim 27,648$ link variables.
- A standard numerical integral (like Simpson's rule) with just 10 points per link would require $10^{27,648}$ evaluations.
- The universe hasn't existed long enough for a supercomputer to do this.

Monte Carlo Pipeline

Goal: Compute observables in SU(2) pure gauge theory.

Monte Carlo Pipeline

1: Construct lattice Λ; initialise links $U_\mu(x) \in SU(2)$
2: **for** $n = 1$ to N_{therm} **do**
3: Perform one full **sweep** (Heat Bath over all links)
4: **end for**
5: **for** $m = 1$ to N_{meas} **do**
6: Perform one full **sweep**
7: Measure: Plaquette, Polyakov Loop, Staples
8: **end for**
9: Compute averages and statistical errors (Jackknife)

What is a Sweep?

A sweep updates every link on the lattice exactly once.

Lattice Sweep

1: **for** each site $x \in \Lambda$ **do**
2: **for** each direction $\mu \in \{0, 1, 2, 3\}$ **do**
3: $V \leftarrow \text{Staple}(x, \mu)$
4: $U_\mu(x) \leftarrow \text{HeatBathUpdate}\big(U_\mu(x), V\big)$
5: **end for**
6: **end for**

A Key Insight

Before describing the Heat Bath update, we need to fix the representation of $SU(2)$.

Storing links as 2×2 complex matrices works, but is **redundant and expensive**.

There is a far more elegant — and computationally cheaper — choice.

SU(2) as Unit Quaternions

Key fact:

$$SU(2) \cong \left\{(a_0, a_1, a_2, a_3) \in \mathbb{R}^4 \;\middle|\; a_0^2 + a_1^2 + a_2^2 + a_3^2 = 1\right\}$$

Matrix representation:

$$U = a_0\,\mathbb{I} + i\sum_{i=1}^{3} a_i\,\sigma_i = \begin{pmatrix} a_0 + ia_3 & a_2 + ia_1 \\ -a_2 + ia_1 & a_0 - ia_3 \end{pmatrix}$$

Interpretation:

- $SU(2)$ group elements $\longleftrightarrow$ points on the 3-sphere S^3.
- Group multiplication $\longleftrightarrow$ quaternion multiplication.
- Group closure is automatic; no reunitarisation needed.

SU(2) Algebra: Elegance of Quaternions

Group Operations

function MULTIPLY(U, V)
    $w_0 \leftarrow u_0 v_0 - u_1 v_1 - u_2 v_2 - u_3 v_3$
    $w_1 \leftarrow u_0 v_1 + u_1 v_0 + u_2 v_3 - u_3 v_2$
    $w_2 \leftarrow u_0 v_2 - u_1 v_3 + u_2 v_0 + u_3 v_1$
    $w_3 \leftarrow u_0 v_3 + u_1 v_2 - u_2 v_1 + u_3 v_0$
    return (w_0, w_1, w_2, w_3)
end function

function DAGGER(U)
    return $(u_0, -u_1, -u_2, -u_3)$
end function

Why Quaternions Win: Computational Comparison

Matrix Form

- 8 real numbers stored
- Unitarity must be enforced: $U^\dagger U = \mathbb{I}$
- Numerical drift $\Rightarrow$ periodic reunitarisation
- Complex-valued multiplications throughout

$\Rightarrow$ ~ 32 real operations per product

Quaternion Form

- 4 real numbers stored
- Constraint: $\sum_i a_i^2 = 1$
- Closure automatically preserved
- Conjugate without inversion: $U^\dagger = (a_0, -\vec{a})$

$\Rightarrow$ ~ 16 real operations per product

$2\times$ fewer operations, $2\times$ less memory — with no loss of precision.

Initialising the Lattice

Initialisation

for each site $x \in \Lambda$ **do**
    for each direction $\mu \in \{0, 1, 2, 3\}$ **do**
        if cold start **then**
            $U_\mu(x) \leftarrow \mathbb{I}$
        else
            $(a_0, a_1, a_2, a_3) \leftarrow$ RandomUnitQuaternion()
            $U_\mu(x) \leftarrow (a_0, a_1, a_2, a_3)$
        end if
    end for
end for

Staple Construction

Compute Staple $V_\mu(x)$

1: $V \leftarrow 0$
2: **for** $\nu \neq \mu$ **do**
3: $\quad F \leftarrow U_\nu(x)\, U_\mu(x+\hat{\nu})\, U_\nu^\dagger(x+\hat{\mu})$
4: $\quad B \leftarrow U_\nu^\dagger(x-\hat{\nu})\, U_\mu(x-\hat{\nu})\, U_\nu(x-\hat{\nu}+\hat{\mu})$
5: $\quad V \leftarrow V + F + B$
6: **end for**
7: **return** V

Polyakov Loop

Loop at fixed $\vec{x}$

1: $L \leftarrow \mathbb{I}$
2: **for** $t = 0$ to $N_t - 1$ **do**
3: $\quad L \leftarrow \text{Multiply}(L, U_0(\vec{x}, t))$
4: **end for**
5: **return** $\frac{1}{2}\,\text{Tr}(L)$

Spatial Average

1: $P \leftarrow 0$
2: **for** each spatial site $\vec{x}$ **do**
3: $\quad P \leftarrow P + \text{Polyakov}(\vec{x})$
4: **end for**
5: **return** P/N_s^3

SU(2) Heat Bath Update

Heat Bath Update

function HeatBathUpdate($U_\mu(x)$, V)
  $a \leftarrow \sqrt{\det(V)}$
  $\tilde{V} \leftarrow V/a$
  $w_0 \leftarrow \text{SampleScalar}(\beta\, a)$
  $\hat{n} \leftarrow \text{RandomUnitVector}()$
  $W \leftarrow \left(w_0,\ \sqrt{1-w_0^2}\,\hat{n}\right)$
  return Multiply$\left(W, \text{Dagger}(\tilde{V})\right)$
end function

Parallelisation: 8-Colour Checkerboard

The hypercubic lattice is bipartite. Each site is assigned a parity based on $(x+y+z+t) \bmod 2$.

- Loop over 4 directions (μ) sequentially on the outside.
- **Pass 1:** Freeze Odd sites; update all Even sites in parallel.
  Even sites read only Odd neighbours $\Rightarrow$ **no race conditions**.
- **Pass 2:** Freeze Even sites; update all Odd sites in parallel.

Checkerboard Sweep

1: **for** $\mu = 0$ to 3 **do**
2: Update all even sites in parallel
3: Synchronise
4: Update all odd sites in parallel
5: Synchronise
6: **end for**

Statistical Errors: Jackknife Estimator

Jackknife Estimator

1: Partition data into N_B bins
2: $\bar{x} \leftarrow$ mean over all bins
3: **for** $i = 1$ to N_B **do**
4: Compute leave-one-out mean $x_{(i)}$ excluding bin i
5: **end for**
6: $\sigma^2 \leftarrow \frac{N_B - 1}{N_B} \sum_i \left(x_{(i)} - \bar{x}\right)^2$
7: **return** $\sqrt{\sigma^2}$

Results

Order Parameter: Polyakov Loop

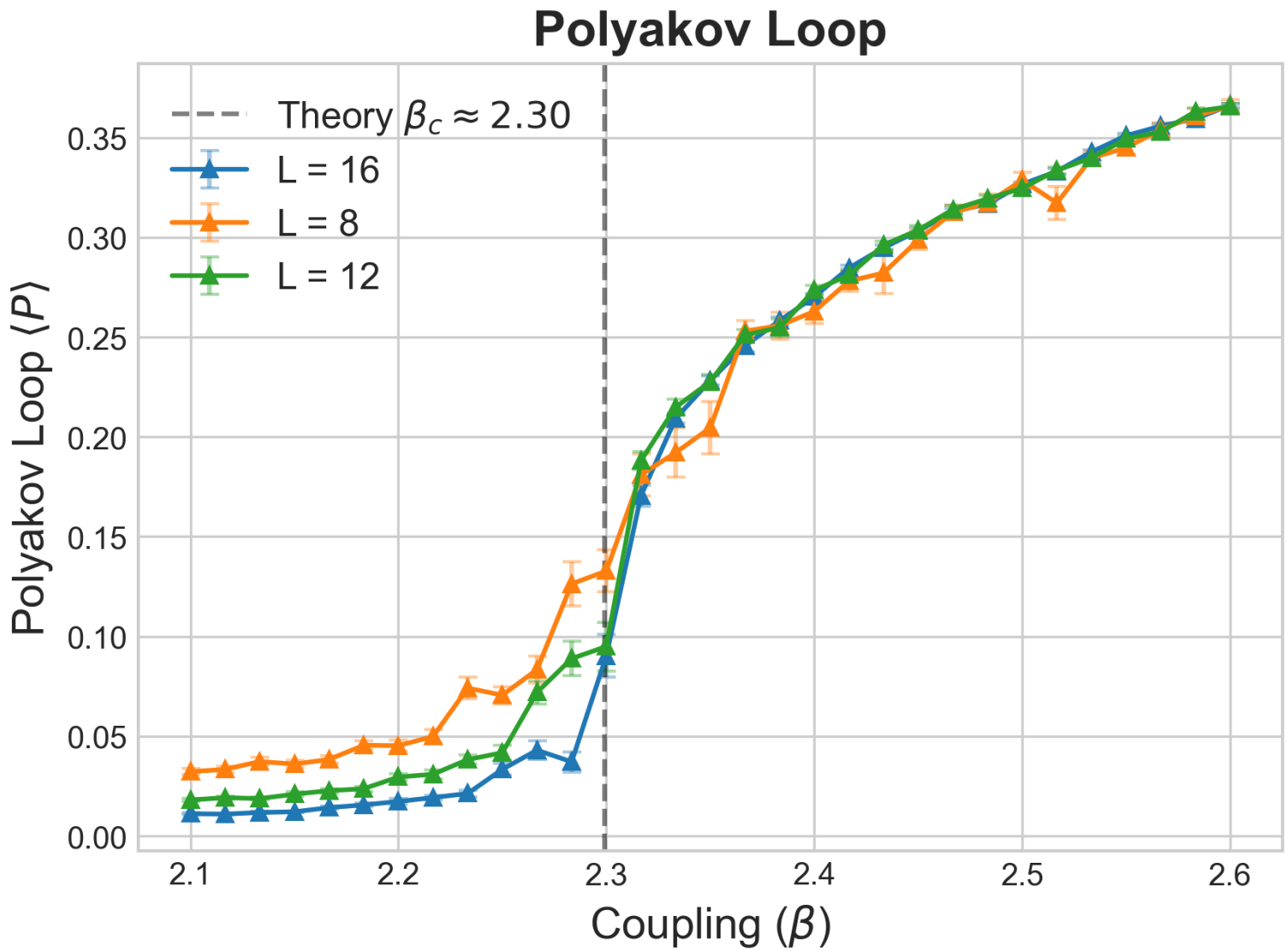


Figure: Polyakov loop $\langle L \rangle$ vs. β for various lattice sizes.

Susceptibility $\chi_L \propto \langle L^2 \rangle - \langle L \rangle^2$

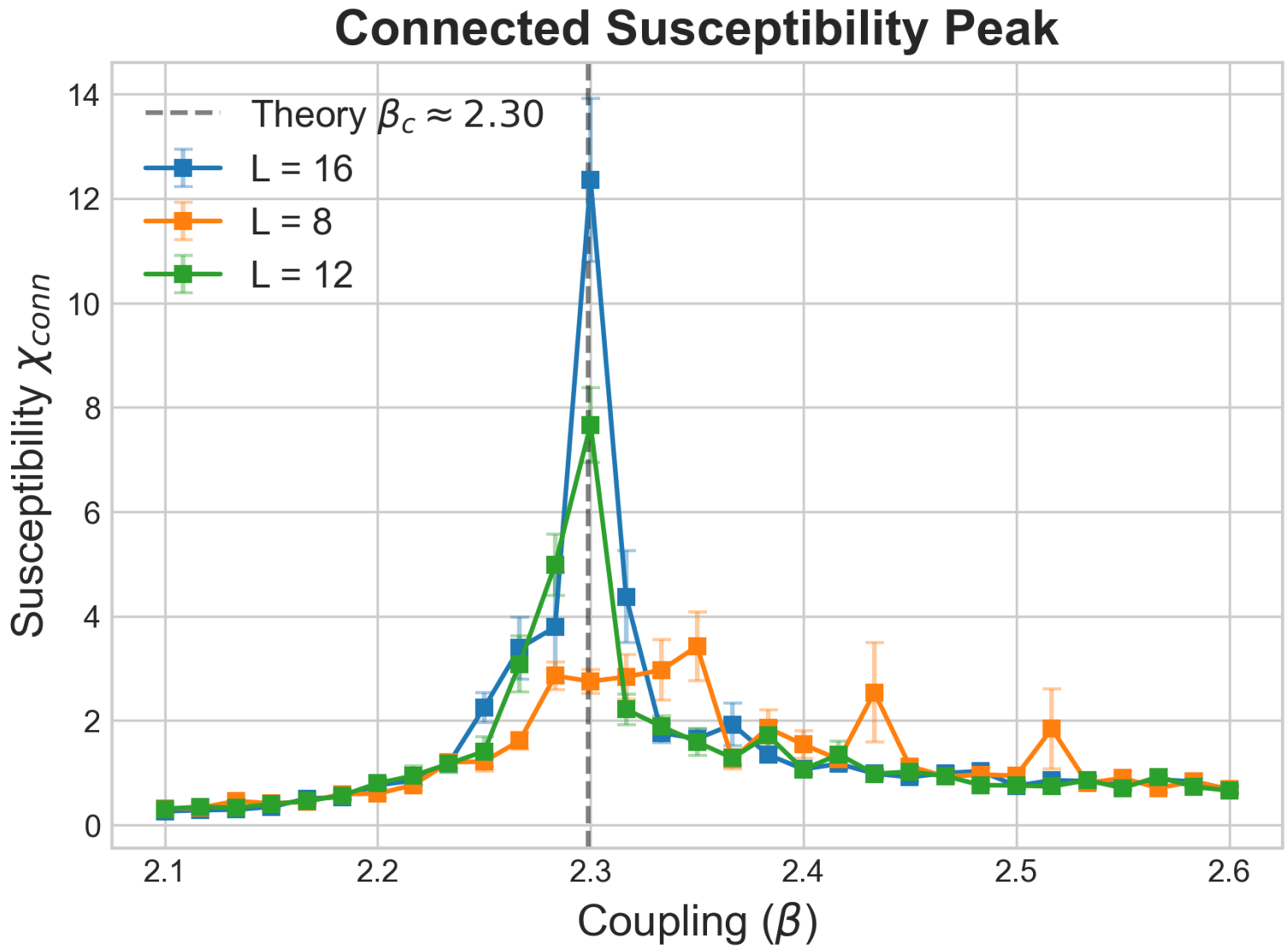


Figure: Susceptibility vs. β for various lattice sizes. Peak sharpens with N_s.

Binder Cumulant $U_4 = 1 - \dfrac{\langle L^4 \rangle}{3\langle L^2 \rangle^2}$

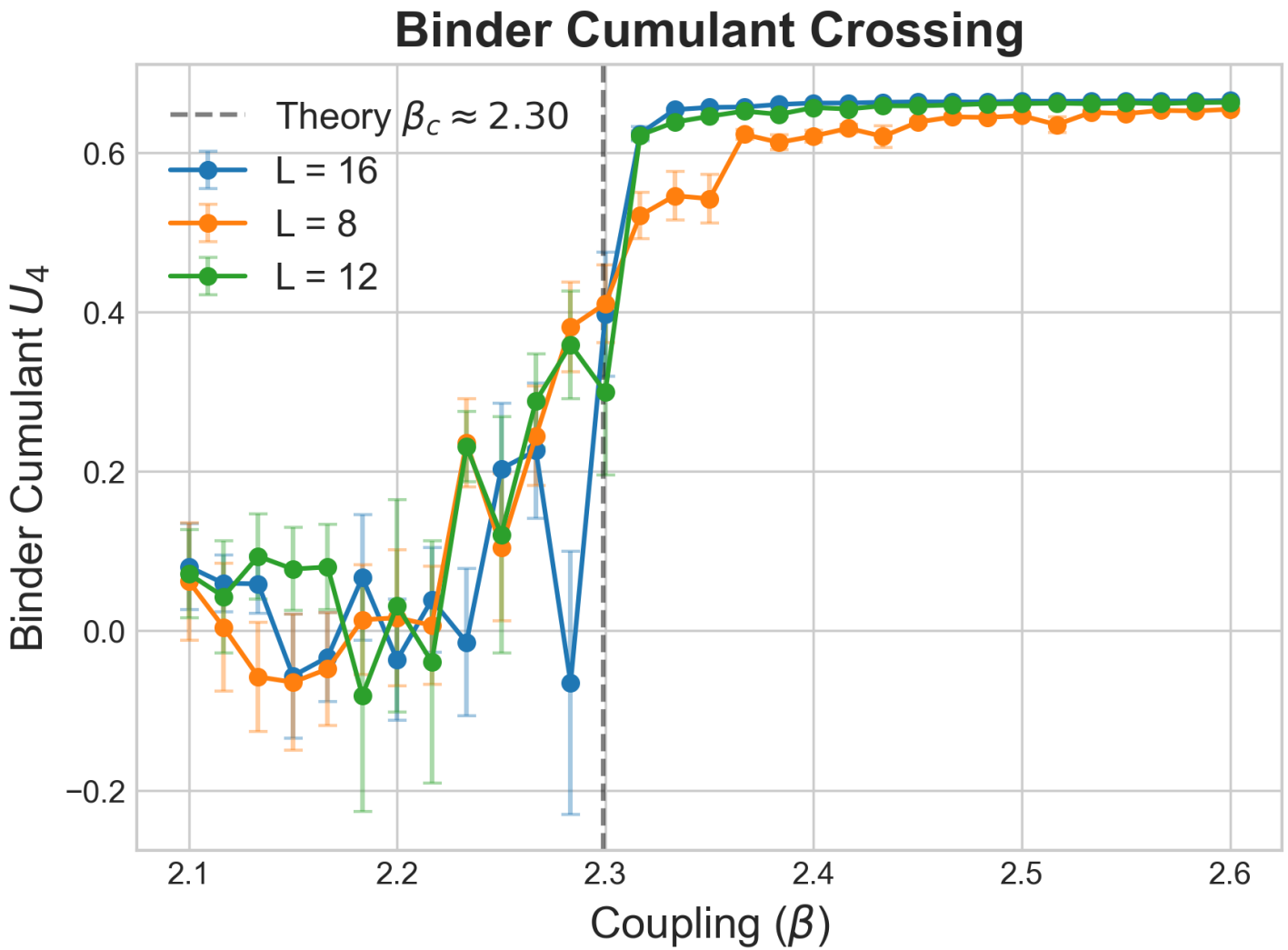


Figure: Binder cumulant vs. β. Crossing point identifies the pseudo-critical β_c.

OpenMP Scaling with Thread Count

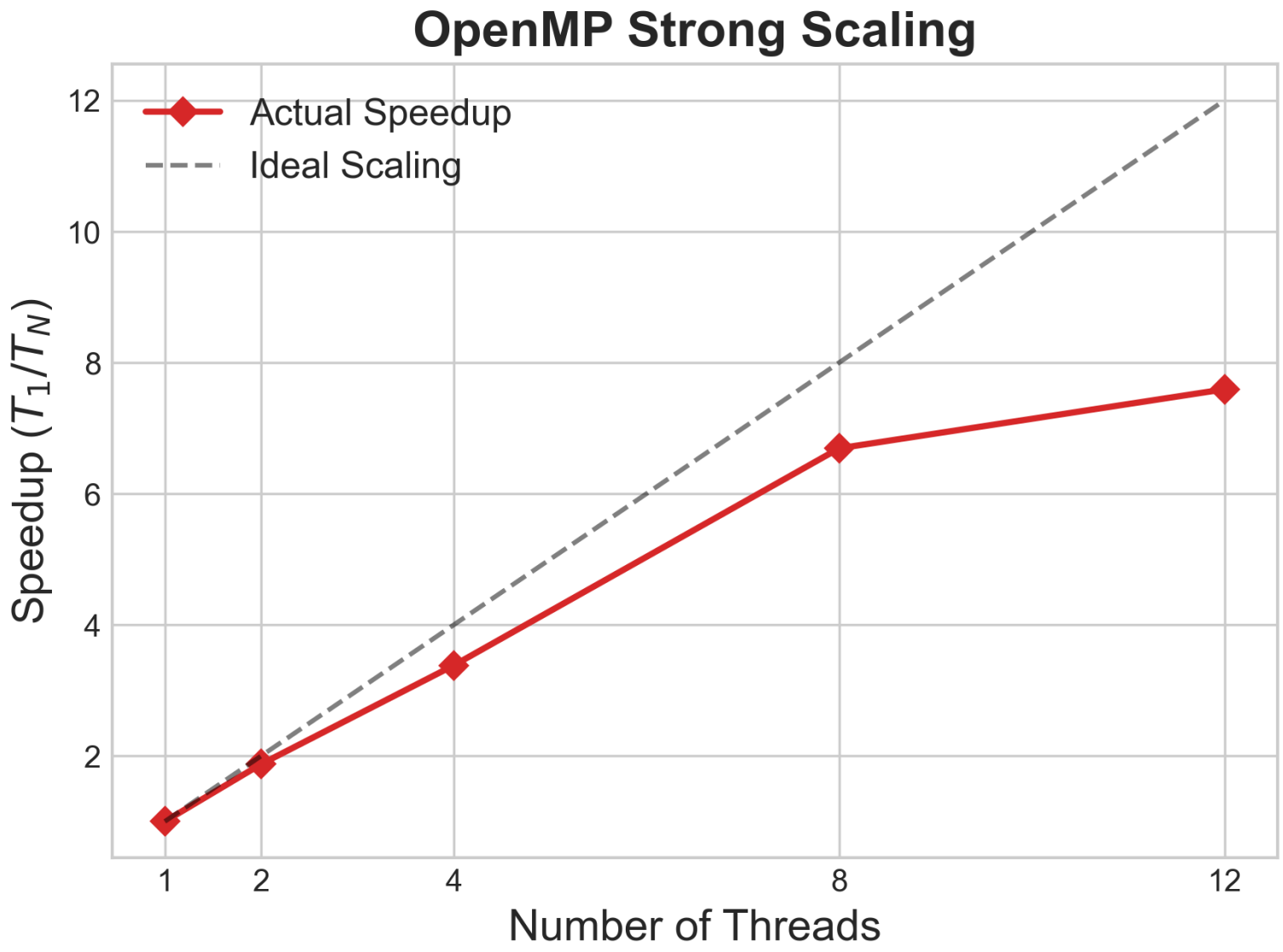


Figure: Speedup vs. number of CPU threads for 500 Heat Bath sweeps.

Conclusion

Summary

What we built:

- Full $SU(2)$ lattice gauge theory simulation in 4D.
- Quaternion representation for exact, efficient group arithmetic.
- Heat Bath algorithm for ergodic sampling of the path integral.
- 8-colour checkerboard parallelisation with OpenMP.
- Jackknife error analysis for correlated Monte Carlo data.

What we found:

- Clear confinement–deconfinement transition as β is varied.
- Susceptibility peak sharpens with lattice size N_s, consistent with a phase transition.
- Binder cumulant crossing fixes the pseudo-critical coupling β_c.
- Near-linear OpenMP speedup up to ~ 8 threads.

Scaling of β_c with N_τ

Definition

The critical coupling β_c depends on the temporal lattice extent N_τ:

$$T = \frac{1}{N_\tau a(\beta)}, \quad T_c \text{ fixed } \Rightarrow \beta_c(N_\tau)$$

Larger $N_\tau \Rightarrow$ smaller lattice spacing $a \Rightarrow$ larger β_c.

Physical Meaning:

- β_c shifts upward with N_τ due to asymptotic freedom.
- Ensures the deconfinement transition occurs at fixed physical T_c.
- Used to extrapolate results to the continuum limit ($a \to 0$).

N_τ	β_c (SU(2))
4	≈ 2.30
6	≈ 2.43
8	≈ 2.51

Future Directions

Physics extensions:

- Extend to full $SU(3)$ (QCD) with quarks.
- Measure Wilson loops and extract the static quark potential $V(r)$.
- Finite-temperature equation of state.

Algorithmic improvements:

- Replace Heat Bath with full **Hamiltonian Monte Carlo** for reduced autocorrelation.
- GPU acceleration for large lattices.

Thank You

Questions?

Credits

Special thanks to ***Dr. Navdeep Dhindsa***, Post Doc @ TIFR and ***Dr. Anosh Joseph***, Assistant Professor @ University of Witwatersand, for providing me with their invaluable guidance.